

Livestock Tracking & Processing System — Developer Handoff Document (V1)

1. Document Purpose

Yeh document development team ke liye banaya gaya hai taake:

- system ka complete flow samajh aaye
 - kis module par kya banana hai clear ho
 - backend aur frontend ka kaam define ho
 - development start karna easy ho
-

2. System Summary (Simple)

Yeh system bakron ko track karega:

👉 entry → branch → slaughter → processing → inventory → report

3. System Architecture (High Level)

Application Type:

👉 Responsive Web App

Suggested Stack:

Frontend:

- React.js (MUI recommended)

Backend:

- Node.js + Express

Database:

- MongoDB
-

Roles:

- Admin
 - Manager
 - Staff
-

4. End-to-End System Flow

Flow 1: Batch Create

Admin:

- batch create karega
 - total animals enter karega
-

Flow 2: Branch Allocation

Admin:

- batch ko branches mein divide karega
-

Flow 3: Branch Receive

Manager:

- receivedAnimals confirm karega
-

Flow 4: Slaughter Entry

Staff:

- daily slaughter count enter karega

System:

- remaining animals calculate karega
-

Flow 5: Processing Entry

Staff:

- gosht weight enter karega
- khal aur paye enter karega

System:

- inventory update karega
-

Flow 6: Reporting

Admin:

- summary reports dekhega
-

5. Module Breakdown (Dev Tasks)



Module 1: Livestock Batch

Backend:

- create batch
- get batches

Frontend:

- batch create form
 - batch list view
-



Module 2: Branch Allocation

Backend:

- assign animals to branch
- update allocation

Frontend:

- allocation screen
 - branch-wise split UI
-



Module 3: Branch Receive

Backend:

- update receivedAnimals

Frontend:

- receive button + input
-



Module 4: Slaughter

Backend:

- create slaughter entry
- validate count

Logic:

remaining = assigned - total slaughter

Frontend:

- simple entry form (count + date)
-



Module 5: Processing

Backend:

- create processing record
- link with slaughter

Logic:

- meatWeight add to inventory
- skins increment
- paye increment

Frontend:

- input form (weight + counts)
-

Module 6: Inventory

Backend:

- maintain stock per branch

Frontend:

- inventory view (table)
-

Module 7: Reporting

Backend:

- aggregate queries

Frontend:

- dashboard cards:
 - total animals
 - slaughtered
 - remaining
 - total meat
-

Module 8: Auth

Backend:

- login API
- role-based middleware

Frontend:

- login page
 - role-based redirect
-

6. Database Collections

Users

```
{  
  name,  
  email,  
  password,  
  role,  
  branchId,  
  status  
}
```

Batch

```
{  
  totalAnimals,  
  category,  
  arrivalDate  
}
```

BranchAllocation

```
{  
  batchId,  
  branchId,  
  assignedAnimals,  
  receivedAnimals  
}
```

```
}
```

Slaughter

```
{  
  branchId,  
  date,  
  count  
}
```

Processing

```
{  
  branchId,  
  slaughterId,  
  meatWeight,  
  skins,  
  paye  
}
```

Inventory

```
{  
  branchId,  
  meatStock,  
  skinStock,  
  payeStock  
}
```

7. Critical Business Logic (MUST IMPLEMENT)

Rule 1:

slaughterCount <= availableAnimals

Rule 2:

processing only allowed after slaughter

Rule 3:

inventory cannot go negative

Rule 4:

branch can only access its own data

8. API Endpoints (Basic)

Batch

- POST /batch
 - GET /batch
-

Allocation

- POST /allocation
 - GET /allocation
-

Slaughter

- POST /slaughter
 - GET /slaughter
-

Processing

- POST /processing
 - GET /processing
-

Inventory

- GET /inventory
-

Auth

- POST /login
-

9. Frontend Screens

Admin

- Dashboard
 - Create Batch
 - Allocation Screen
 - Reports
-

Manager

- Receive Screen
 - Slaughter Screen
 - Processing Screen
-

Staff

- Quick Entry Form

10. 🧠 Development Order (IMPORTANT)

Phase 1:

- Auth
 - Batch
 - Allocation
-

Phase 2:

- Slaughter
 - Processing
-

Phase 3:

- Inventory
 - Reports
-

11. 🚀 MVP Rules

MUST WORK:

- batch entry
 - slaughter tracking
 - processing entry
 - inventory
-

DON'T BUILD NOW:

- mobile app
 - barcode system
 - AI features
-

12. Developer Notes

- Keep UI simple (forms + tables)
 - Focus on logic correctness
 - Avoid over-engineering
 - Use validation everywhere
 - Test edge cases (over slaughter, wrong inputs)
-

13. Definition of Done

System complete tab mana jayega jab:

- admin batch create kar sake
- branch allocation ho jaye
- slaughter track ho
- processing entry ho
- inventory auto update ho
- reports visible ho